MIL-R-24085A(SHIPS)
18 March 1968
SUPERSEDING
MIL-R-24085(SHIPS)
30 September 1964
(See 6.4)

MILITARY SPECIFICATION

# REFRIGERATING UNIT, CENTRIFUGAL FOR AIR CONDITIONING

## 1. SCOPE

1.1 Scope. This specification covers centrifugal refrigeration unit with integrated water chiller for use in an environmental control system for surface ships or submarines.

# 1.2 Classification. The refrigerating unit shall be of the following types, classes and arrangements as specified (see 6.1).

Type I - surface ships
Type II - submarines
Class 1 - using hermetic compressor
Class 2 - using open compressor
Arrangement A - right hand
Arrangement B - left hand
Arrangement C - double unit
Arrangement D - right hand side by side
Arrangement E - left hand side by side

## 2. APPLICABLE DOCUMENTS

# 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-N-281 - Nickel-Copper-Alloy, Bar, Plate, Rod, Sheet, Strip, Wire, Forgings, and Structural And Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought, (K-Monel).
TT-P-645 - Primer, Paint, Zinc-Chromate, Alkyd Type.
WW-T-799 - Tube, Copper, Seamless (For Use With Solder-Type or Flared Tube Fittings).

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein) Packaging of.
MIL-P-116 - Preservation, Methods of.
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
MIL-S-901 - Shock Tests H.I. (High Impact), Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-1000 - Drawings, Engineering and Associated Lists.
MIL-D-1000/2 - Drawings, Engineering and Associated Lists.
MIL-C-2212 - Controllers, Alternating-Current, Naval Shipboard.
MIL-M-9868 - Microfilming of Engineering Documents, 35MM, Requirements for.
MIL-C-9877 - Cards, Aperture.
MIL-P-15024 - Plates, Identification--Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-E-15090 - Enamel, Equipment, Light Gray (Formula No. 11).
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-C-15328 - Primer Pretreatment (Formula No. 117 for Metals).
MIL-H-15424 - Hand Tools, Packaging of.
MIL-C-15726 - Copper-Nickel Alloy Rod, Flat Products (Flat Wire, Strip, Sheet, Bar and Plate) and Forgings.
MIL-C-15730 - Coolers, Fluid, Industrial, Naval Shipboard, Lubricating Oil, Hydraulic Oil, and Fresh Water.
MIL-R-16743 - Refrigerating Systems, Mechanical; and Refrigeration System Components -- Dichlorodifluoromethane, Type 12 Naval Shipboard.
MIL-M-17059 - Motors, Alternating-Current, Fractional HP (Shipboard Use).



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MIL-M-17060 - Motors; 60 Cycles, Alternating Current, Integral, HP (Shipboard Use).
MIL-P-17545 - Primer Coating, Steel Ship Maintenance, Alkyd-Red Lead Type, Formula No. 116.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-C-17728 - Controls and Drive Assembly, Tube Expander, Automatic.
MIL-I-18997 - Indicator, Pressure, Panel Mounted or Case Supported, General Specification.
MIL-S-19500 - Semi-Conductor Devices, General Specification for.
MIL-A-19521 - Anodes, Corrosion Preventive, Zinc, and Plugs, Zinc Anode Retaining: Design of and Installation in Shipboard Condensers and Heat Exchangers.
MIL-T-19646 - Thermometers, Remote Reading, Self-Indicating Dial, Gas Actuated.
MIL-F-20042 - Flanges, Pipe, Bronze (Silver Brazing).
MIL-C-20159 - Copper-Nickel Alloy (70-30 and 90-10), Castings.
MIL-T-22214 - Tubes, 70-30 and 90-10 Copper-Nickel Alloy, Condenser And Heat Exchanger with Integral Fins.
MIL-B-23921 - Bronze, Nickel Aluminum Castings for Seawater Service.
MIL-I-24137 - Iron Castings, Nodular Graphitic (Ductile Iron) and Nodular Graphitic (Corrosion Resisting, Austenitic; Low Mangetic Permeability) (For Shipboard Application).
MIL-W-24270 - Wells for Indicators or Thermal Elements: General Specification.
MIL-M-38761 - Microfilming and Photographing of Engineering/Technical Data and Related Documents: PCAM Card Preparation, Engineering Data Micro-Reproduction System, General Requirements For, Preparation of.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

MILITARY

MIL-STD-101 - Color Code for Pipelines and for Compressed-Gas Cylinders.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-195 - Marking of Connections for Electric Assemblies.
MIL-STD-278 - Fabrication, Welding and Inspection of Machinery, Piping and Pressure Vessels for Ships of the United States Navy.
MIL-STD-290 - Packaging, Packing and Marking of Petroleum and Related Products.
MIL-STD-438 - Schedule of Piping, Valves, Fittings and Associated Piping Components for Submarine Service.
MIL-STD-701 - Lists of Standard Semiconductor Devices.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-758 - Packaging, Procedures for Submarine Repair Parts Utilizing Transparent, Flexible, Heat Sealable Film.
MIL-STD-769 - Thermal Insulation Requirements for Machinery and Piping.
MIL-STD-777 - Schedule of Piping, Valves, Fittings and Associated Piping Components for Surface Ships.
MIL-STD-804 - Formats and Coding of Aperture, Copy and Tabulating Cards for Engineering Data, Micro-Reproduction System.

PUBLICATIONS

NAVSHIPS

0900-001-7000 - Fabrication and Inspection of Brazed Piping Systems.
SDB-63 - Structural Design Basis.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

# 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.



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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A105-65 - Forged or Rolled Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service.
A204-66 - Molybdenum Alloy Steel Plates for Pressure Vessels.
A266-67 - Carbon Steel Seamless Drum Forgings.
A515-67 - Carbon Steel Plates for Pressure Vessels for Intermediate and Higher Temperature Service.
A536-67 - Ductile Iron Castings.
B61-63 - Steam or Valve Bronze Castings.
B98-66 - Copper-Silicon Alloy Rod, Bar and Shapes.
B139-66 - Phosphor Bronze Rod, Bar, and Shapes.
B169-66 - Aluminium Bronze Plate, Sheet, Strip and Rolled Bar.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

UNITED STATES OF AMERICA STANDARDS INSTITUTE (USAS)
B46.1-1962 - Surface Texture.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, N. Y. 10016.)

WELDING RESEARCH COUNCIL
Bulletin 107 - Local Stresses in Spherical and Cylindrical Shells Due to External Loadings.

(Application for copies should be addressed to the Welding Research Council, 345 East 47th Street, New York, N. Y. 10017.)

NATIONAL MOTOR FREIGHT CLASSIFICATION
Classes and Rules

(Application for copies should be addressed to the Issuing Officer, National Motor Freight, 1616 P Street, N. W., Washington, D. C. 20036.)

DEPARTMENT OF TRANSPORTATION
Interstate Commerce Commission Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Boiler and Pressure Vessel Code, Section III

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

## 3. REQUIREMENTS

### 3.1 General shipboard design conditions.

# 3.1.1 The equipment shall perform in accordance with the requirements specified herein under the following environmental design conditions, as applicable:

| | | Minimum | Maximum |
|---|---|---|---|
| (a) | Sea water temperature range (type I) | 28°F. | 88°F. |
| (b) | Sea water temperature range (type II) | 28°F. | 85°F. |
| (c) | Machinery space temperature | 40°F. | 122°F. |
| (d) | Surface ship sea water pressure pounds per square inch gage, (psig) | 35 | 180 |
| (e) | Submarine sea water pressure | As specified | (see 6.1) |



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| | | Minimum | Maximum |
|---|---|---|---|
| (f) | Submarine atmosphere, inches Hg absolute | 24 | 38 |
| (g) | Chilled water pressure, psig | -- | 125 |

Type II units shall not be damaged when subjected to an ambient pressure between 10 and 30 pounds per square inch absolute (psia).

3.1.2 The equipment shall be designed to operate satisfactorily, to maintain adequate lubrication, to avoid loss of oil or fluids and to drain without spillage under the following conditions:

(a) Type I units:

(1) Up to 15 degrees from the normal horizontal position in any direction (permanently inclined).
(2) With the ship rolling up to 45 degrees from the vertical to either side.
(3) With the ship pitching 10 degrees up and down from the normal horizontal plane.

(b) Type II units:

| | | |
|---|---|---|
| (1) | Trim (permanent) | 30 degrees |
| (2) | Pitching | 10 degrees |
| (3) | Time of complete pitch cycle | 9 seconds |
| (4) | List (permanent) | 15 degrees |
| (5) | Rolling | 60 degrees |
| (6) | Time of complete roll cycle | 15 seconds |

NOTE: 1. Degrees indicated are measured from the vertical to either side for list and roll, and from the normal horizontal plane up or down for trim and pitch.
2. Machinery shall not be damaged due to trim angles up to 45 degrees.

(c) The conditions resulting from the following combinations shall be met for submarines:
(1) List and trim.
(2) List and pitch.
(3) Roll and trim.
(4) Roll and pitch.

# 3.2 General design of equipment. The unit shall be designed to chill circulating water for air conditioning application by use of a centrifugal compressor utilizing R-11 as the refrigerant unless otherwise specified (see 6.1). Where refrigerant other than R-11 is specified the pressure ranges of gages, switches and pressure tests shall be compatible with the refrigerant specified.

# 3.2.1 Operating requirements. Each centrifugal unit shall have a capacity of not less than the number of tons of refrigeration specified (see 6.1) under the following conditions:

(a) Minimum refrigerant temperature - 33°F.
(b) Varying entering condensing sea water temperature (maximum) - 88°F. (type I), 85°F. (type II).
(c) Condensing temperature (maximum) - 105°F.
(d) Outlet chilled water temperature as specified (see 6.1).
(e) Chilled water flow - 3.6 gallons per minute (gpm) per ton (approximate) unless otherwise specified (see 6.1)[1/].
(f) Chilled water pressure drop through chiller - 10 pounds per square inch (psi) (maximum) at condition (e) above[1/].
(g) Chilled water velocity - 10 feet per second (fps) (maximum) at condition (e) above[1/].
(h) Sea water velocity through tubes - 6.0 fps (maximum).
(i) Sea water velocity through nozzles - 7.5 fps (maximum).

[1/] In addition each unit shall produce the capacity specified (see 6.1) plus or minus 5 percent when: the chilled water flow is 4.5 gpm per ton (approximate); the chilled water pressure drop through the chiller is 15 psi (maximum, the chilled water velocity is 12.5 fps (maximum) and all other conditions the same as indicated above.



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(j) Condenser waterside fouling coefficient - 0.0005 scale factor.
(k) Sea water flow - 4 gpm per ton maximum unless otherwise specified (see 6.1).
(l) Sea water pressure drop through condenser - 10 psi (maximum).

# 3.2.1.1 Unless specific Naval Ship Engineering Center (NAVSEC) (see 6.2), approval is obtained the capacity of the unit specified (see 6.1) shall agree with one of the following standard capacities and all of the centrifugal air conditioning units installed on a ship shall be of the same capacity and same design.

Standard capacities in tons of refrigeration

| | | |
|---|---|---|
| 150 | 250 | 400 |
| 175 | 300 | 450 |
| 200 | 350 | 500 |

# 3.3 General assembly. Each unit shall consist of a centrifugal compressor, speed increasing gears (if necessary), electric motor drive, condenser, and chiller all factory mounted as a single unit to meet the requirements of the intended service. The unit shall also include safety and operating controls, control panel(s), purge unit or combination purge-evacuation unit as required in 3.4.3 and all necessary interconnecting piping, (excluding interconnecting lines to remotely installed support equipment) valves, fittings, accessories, instruments, indicating devices and motor controllers necessary for operation.

# 3.3.1 Initial charge of refrigerant shall be shipped in 100 pound commercial containers. Initial charge of oil shall be fresh oil charged into compressor oil reservoir at conclusion of all required tests after oil used during tests is thoroughly drained. No additional initial oil charge in separate containers shall be required.

3.3.2 The unit should be designed and fabricated so it can be easily maintained, disassembled and repaired. The contractor shall demonstrate these features of the unit (see 4.4.5).

# 3.3.3 Means shall be provided to permit cleaning the condenser and chiller by flushing with refrigerant or other cleaning agent. The condenser and chiller shells shall have connections for this purpose, those in the chiller shall be located at the lowest point on the chiller shell.

# 3.4 Arrangement. Where space is critical the arrangement of the complete assembly shall not exceed the dimensions specified (see 6.1). This space must include the base or foundation support plates and all piping and valves (except water regulating valve) which are a part of the unit.

# 3.4.1 The compressor and motor shafts shall be parallel to the condenser and chiller and arranged A, B, C, D or E as shown on figures 1, 2 and 3 and specified (see 6.1).

# 3.4.2 Where arrangement C is specified, it shall require two separate and completely independent units of equal capacity to be mounted as a single assembly.

# 3.4.3 One purge unit and vacuum pump, for type I units, or one combination purge-evacuation unit, for type II units, shall be provided for each unit unless otherwise specified (see 6.1).

# 3.4.4 The unit shall be a right hand or a left hand arrangement with respect to location of instruments and controls. The right hand arrangement shall have the gageboard on the right side of the assembly when facing the condenser water inlet connection. Dehydrators, valves, and controls not mounted on the gageboard or not accessible for servicing from either end of the assembly shall be accessible from the right side. The sight glasses shall be visible from the right side. The left hand assembly shall be opposite the right hand assembly, the gageboard shall be on the left side of the unit and the components as indicated above shall be accessible or visible from the left side of the assembly.

# 3.4.5 The sketches of arrangements A, B, C, D and E contained herein are included to illustrate examples of these arrangements. They are not meant to dictate the relative location or design of the components which make up the air conditioning unit.



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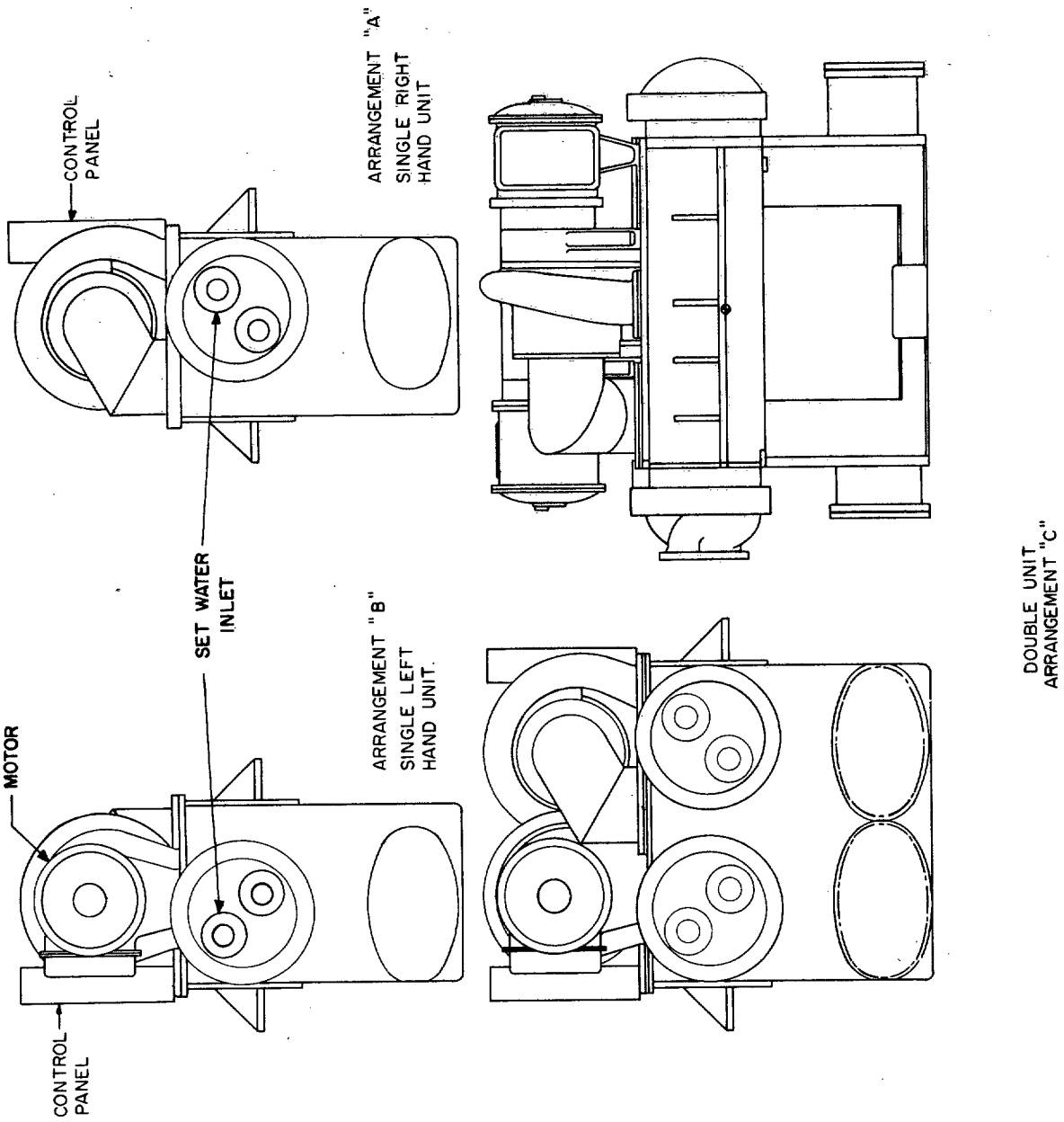



Figure 1 - Unit Arrangement



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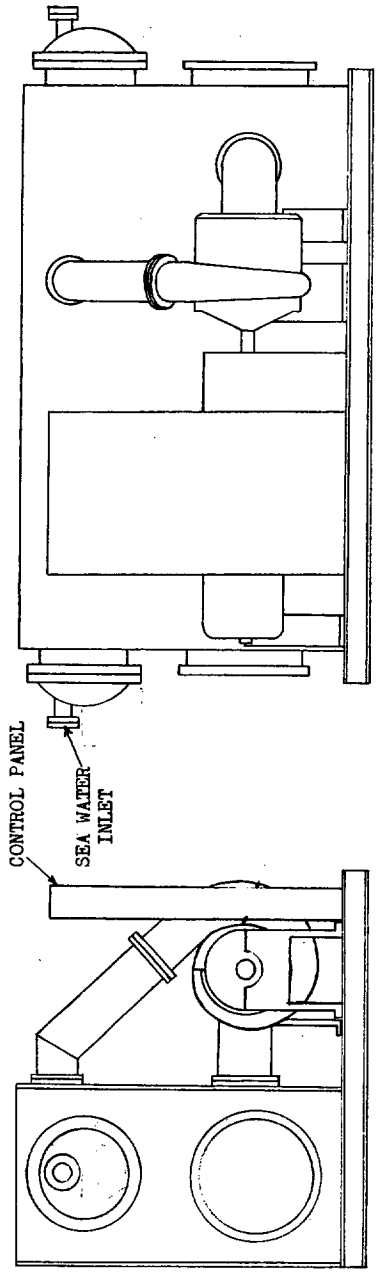
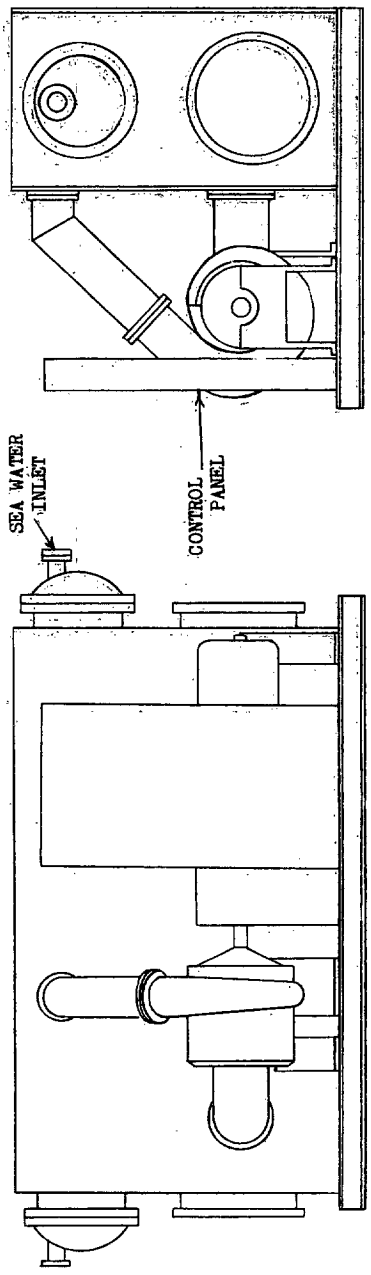



Figure 2 - Arrangement D-right hand unit (side by side).



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Figure 3 - Arrangement E-left hand unit (side by side).



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3.5 Reliability assurance.

3.5.1 The principle of maximum reliability is paramount and no compromise of this principle shall be made with any other basic requirement of design. Where wear or corrosion is unavoidable, the parts subjected to such wear or corrosion shall be of the best material for the purpose in order to reduce those detrimental effects to a minimum. It is the intention of this specification to obtain equipment of such design that it will have an operating life of at least 40,000 hours. The basis for design of replaceable parts shall be an equivalent of three years of ship operation (approximately 10,000 hours) before replacement is necessary.

3.5.1.1 The designer should take cognizance of the conditions under which equipment will be maintained and repaired on shipboard, and of the fact that the personnel responsible for maintenance and repair may not be seasoned mechanics. Human engineering should be considered in design to preclude or minimize possibility of failure through improper operation or poor maintenance.

3.5.2 The contractor shall establish and maintain an effective reliability assurance program including the following:

(a) Design reviews. The reliability assurance program shall include provisions for the reliability review and evaluation of design as an integral part of the contractor's engineering design procedures. Design or engineering changes occurring during development or production shall be subjected to comparable review procedures.

(b) Production control and monitoring. The reliability assurance program shall provide an economical and effective system of production control and monitoring to assure that reliability achieved in design is maintained during production (see 4.1.1).

(c) Subcontractor and vendor reliability. The reliability assurance program shall include provisions to assure subcontractor and vendor selection and performance consistent with the reliability requirements of the contract and applicable portions of this specification.

(d) Reliability analysis. The contractor shall analyze those factors affecting reliability. The reliability analysis shall include, but shall not be limited to, the following:

(1) List of those parts which experience and judgment show are subject to wear, material deterioration, and service failures.

(2) Specific design features employed to attain the required service life of the parts with due consideration of shipboard environment and resultant conditions. Some suggested design features are choice of materials, compatibility of materials, hardness, surface finishes, fits, clearances, fastenings, equipment protection fail-safe features, reparability and accessibility.

(3) Show by calculation or other means that the design does in fact fulfill the design requirements with the criteria chosen.

(4) Preventive maintenance and servicing requirements necessary to the achievement of reliable equipment. Any unusual steps or precautions necessary in carrying out maintenance and servicing requirements shall be pointed out.

(e) Failure reporting, analysis, and feedback. The reliability assurance program shall incorporate a formalized system for recording, collecting, and analyzing all failures that occur during all testing, installation and operation through the tenure of the contract. Analysis shall be fed back to contractor's engineering, management, and production activities on a timely basis. Failure reports received from using activity shall be integrated into this program for trouble analysis and for experience considerations for future design review.

# 3.6 Materials. The materials of construction shall be similar or equal to the applicable specifications specified herein and shall be the same as successfully used in commercial application except that cast iron where used shall be ductile (nodular) type, in accordance with MIL-I-24137 or grade 65-45-12 in accordance with ASTM A-536.

# 3.6.1 Mercury and mercury containing instruments shall not be used where copper or copper-nickel alloy components may be subjected to direct or indirect mercury contamination.



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# 3.6.1.1 Mercury and mercury containing instruments (except fluorescent or necessary vapor lighting) shall not be used in the manufacture or testing of any equipment destined for installation on a nuclear powered ship. The test reports for this equipment shall contain a certification signed by a responsible official of the manufacturer that no mercury or mercury containing instruments have been used in the manufacture or testing of the equipment.

3.6.2 To prevent destructive electrolysis, direct contact of electrolytically dissimilar materials shall be avoided as far as practicable.

3.6.3 Materials which are capable of producing dangerous gases or other harmful effects under conditions (including fire) encountered in Naval shipboard service shall not be used without specific approval. Magnesium and its alloys shall not be used.

# 3.7 Castings. Castings from materials other than nodular iron shall comply with the requirements of MIL-STD-278. The supplier shall submit for review and approval casting plans detailing the area to be radiographed, the shots to be taken and the acceptance standards to be used. Castings having excessive porosity, sponginess, or cracks will be cause for rejection. Peening of such castings to repair defects will not be permitted. No welding or patching shall be done on iron castings.

3.8 Welding and brazing.

# 3.8.1 Welding. Welding procedures, materials and qualification of welders shall be in accordance with MIL-STD-278.

3.8.2 Brazing. Brazing shall be restricted to piping systems and joints having operating temperatures not higher than 425°F. Silver brazed joints shall be in accordance with NAVSHIPS 0900-001-7000 except that fittings for refrigerant service shall not require pre-inserted solder rings.

3.9 Machine finishes. The following is a general guide to the parts of the machinery which shall be machine finished:

(a) Bearing, mounting, or faying surfaces of machinery bases and foundations which require accurate alignment.
(b) Bearing surfaces for nuts and bolt heads.
(c) Faying surfaces of all projections from the bodies of pedestals, blocks, or other supports meeting finished parts.
(d) Sealing surfaces.
(e) Working parts.

# 3.9.1 Surface finish identification shall be in accordance with USAS B46.1.

3.10 Machine alignment. Foundation feet, brackets, and pads shall be designed so that alignment will be maintained during motion disturbance by shock or vibration.

3.11 Interchangeability. Similar parts including repair parts or corresponding apparatus furnished on the same contract or order or built to same drawings shall be interchangeable without the necessity of further machining or hand fitting or any kind.

3.12 Noise level. The type II equipment shall be tested for airborne and structure-borne noise as required in 4.4.1. The noise limit shall be as specified (see 6.1).

3.13 Shock and vibration.

3.13.1 The design of the complete refrigerating unit and all components shall be such that they are capable of passing the high impact shock tests specified in MIL-S-901 for grade A, class III equipment.

3.13.2 The design information given in Section 6 of MIL-S-901 should be carefully considered in design of equipment. This information is also used by the drawing review activity as a factor in approval of drawings required for production of equipment.

# 3.13.3 The design of equipment and assembly shall be such that no damage will occur or malfuction be caused either by internally excited vibrations, or by the environmental vibrations specified in MIL-STD-167 for frequencies up to and including 33 cycles per second (cps).



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# 3.13.4 Resilient mounts may be used to install equipment where noise attenuation is required, however, command or agency (see 6.3) approval of the mount and its proposed application must be obtained. Resilient mounts shall not be used on equipment where noise attenuation is not required unless command or agency approval of the mount, its proposed application and reasons for using mounts is obtained prior to manufacturing the equipment.

# 3.13.5 Resilient mounts, where required, shall be furnished by the contractor unless otherwise specified (see 6.1).

3.14 Piping, valves and piping connections.

3.14.1 All valves, flanges and fittings for pipe connections shall conform to MIL-STD-777 for type I units and to MIL-STD-438 for type II units.

# 3.14.2 Valve bodies shall be marked to show manufacturer's name, trademark and valve size. If the body is too small a handwheel disc containing the manufacturer's name and valve size shall be supplied.

3.14.3 Tapered pipe threaded connections between piping, machinery and valves, and in piping system joints, are not permitted except by specific approval of NAVSEC.

3.15 Compressor.

# 3.15.1 Standard product. Except as modified herein, the compressor shall be of a type and capacity which has had extensive commercial and industrial application as manufactured by the bidder.

# 3.15.2 The compressor shall be of the centrifugal type. The compressor rotor and impellers shall be statically and dynamically balanced. The shaft(s) shall be fabricated from steel of a grade or type suitable for the application. All bearings shall be replaceable.

3.15.2.1 The compressor and motor shall be either capable of rotating in the reverse direction without incurring any damage, or contain devices or controls which will prevent the compressor and motor from rotating in the reverse direction.

# 3.15.3 The lubricating system shall be of the forced feed type with a pump supplying oil under pressure to the main bearing and thrust bearings (including the compressor motor bearings for class 1 units). An electrically operated oil pump shall be provided with each unit. This oil pump shall be interconnected to the compressor control circuit so that the oil pump will operate for a sufficient time after compressor shutdown to supply lubricating oil until the compressor rotation has completely stopped. In addition the oil pump and compressor control circuit shall be interconnected so the compressor will not start until the oil pump has started and lubricating oil is being circulated to the compressor. The oil pump and lubricating system shall be arranged so that lubricating oil can be circulated for an indefinite period without running the compressor.

# 3.15.3.1 The compressor shall have a single oil reservoir where all of the oil, except that trapped in filters, coolers or other parts of the lubricating system, will return during shutdown. Oil heater(s) shall be installed in the oil reservoir. The heater(s) shall be thermostatically controlled to maintain an oil temperature within the reservoir of 140° to 150°F. or above. The capacity of the electric heaters shall be sufficient to maintain the oil at 140° to 150°F. during shutdown when the machinery space temperature is 40°F.

# 3.15.3.2 The lubricating system shall include an oil cooler using chilled water and oil filters. In addition an oil reclaiming system shall be provided to return oil to the oil reservoir from the liquid refrigerant in the chiller.

3.15.3.2.1 Lubricating oil coolers. Oil coolers may be a separate unit or built into the centrifugal machine. Where the oil cooler is a separate unit, the materials of construction shall comply with MIL-C-15730, type A or B, class 4. Where the unit is a part of the centrifugal unit the water passages shall be 90-10 composition copper-nickel. An automatically operated valve shall be provided to prevent water flow through the cooler when the unit is not in operation.

3.15.3.2.2 Oil filters. Two oil filters arranged in parallel shall be provided. They shall be externally mounted and valved so either filter can be removed and replaced without stopping the unit.



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# 3.15.4 Class 1-hermetic centrifugal compressor. The hermetic centrifugal compressor shall consist of a single or dual stage compressor with its motor drive and speed increasing gears in a totally enclosed unitized assembly so as to operate in the refrigerant atmosphere.

3.15.5 Class 2-open centrifugal compressor. The open centrifugal compressor shall consist of a single or dual stage compressor connected to separate motor through a flexible coupling. The speed increaser gears may be a part of the compressor or a separate gear set.

# 3.15.5.1 Provisions shall be made for access to the compressor shaft seal without the necessity of moving the speed increaser or motor. Provisions shall also be made to permit opening of the compressor where necessary for inspection and repair.

# 3.15.5.2 Gear sets if supplied in separate housings shall comply with MIL-G-17859 as applicable. Gear sets if integral with compressor shall be of a design proven satisfactory for the service intended.

# 3.15.5.3 Gears and bearings shall be lubricated by forced-feed system, having suitable pump, strainer or filter and if supplied in separate gear housing, shall be provided with separate forced-feed lubricating system including oil cooler using chilled water.

# 3.16 Electrical requirements. Motors shall operate from 440 volts, 3 phase, 60 cycle alternating current with a maximum voltage variation of plus or minus 10 percent. Oil heaters and controls may be operated from 115 volt, where desirable, provided integrated control is furnished.

3.16.1 Electrical equipment, components, controls and devices shall conform to the general requirements of MIL-E-917. Vacuum tubes will not be permitted. Permanent markers for wire and cable shall be provided as required by MIL-STD-195.

# 3.16.2 Compressor motor. The compressor motor shall be in accordance with MIL-M-17060, as applicable and as follows:

| | Class 1 | Class 2 |
|---|---|---|
| Service | - - - - | A (unless otherwise specified (see 6.1)) |
| Duty | Continuous | Continuous |
| Enclosure | Hermetic | Dripproof |
| Ambient temperature | - - - - | 50°C |
| Insulation | Compatible with refrigerant being used | B |

# 3.16.3 Controller. The compressor motor controller shall be in accordance with MIL-C-2212 and the following requirements:

(a) Three phase, reduced voltage, closed transition autotransformer or wye delta unless otherwise specified (see 6.1).
(b) Interlock with water pumps.
(c) Dripproof enclosure.
(d) Ambient temperature of 50°C.
(e) Continuous duty.
(f) Magnetic operation.
(g) Low voltage and overload protection.

3.16.3.1 The condenser and chiller circulating water pump motor controllers, compressor motor controller and other required controls shall be electrically interconnected so that the compressor motor cannot be started without condenser and chiller water supply and will stop when water flow fails.

# 3.16.3.2 The water pump interlocks may be installed in the air-conditioning unit control panel or pump motor controller instead of in the compressor motor controller.



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# 3.16.4 Motors for auxiliary equipment. Motors for auxiliary equipment shall be in accordance with MIL-M-17059 or MIL-M-17060 as applicable and shall have the following characteristics:

(a) Service - A (unless otherwise specified (see 6.1)).
(b) Ambient temperature - 50°C.
(c) Duty - Continuous.
(d) Enclosure - Dripproof protected, horizontal or vertical.
(e) Horsepower - As required but not less than 1/4 horsepower.
(f) Revolutions per minute - as required.
(g) Squirrel cage - induction.
(h) Bearings ball.
(i) Design - B.
(j) Insulation - class B.

# 3.16.4.1 Motors for auxiliary equipment shall be sized for effective performance of its intended function in an ambient temperature, as specified in 3.1.2.

3.16.4.2 Hermetic oil pumps may be used with class 1 units only.

# 3.17 Sea water containing components (type II units). Sea water containing components, including valves, piping, water boxes and double tube sheet complex shall be designed for varying submergence conditions. Analysis shall be in accordance with SDB-63 except for calculations of local stresses in spherical and cylindrical shells from external loads. For external loading analysis, Welding Research Council Bulletin 107 shall be used. The cyclic nature of the pressure and thermal loading shall be incorporated in the analysis.

# 3.17.1 The design of water boxes for submarines, shall take into account the cyclic nature of the pressure, sea water temperature, and external force loading.

# 3.17.1.1 In the design of submarine condenser water boxes, the loads exerted by circulating water inlet and discharge piping shall be calculated from I-480 of SDB-63.

# 3.17.2 The structure composed of outer and inner tube sheets together with the short sections of tubes between paired inner and outer sheets shall be designed to withstand the specified design pressure in combination with the specified piping loads with full consideration being given to the cyclic nature of the pressure loading and sea water and refrigerant side temperature.

# 3.17.2.1 The double tube sheet structure shall be capable of withstanding the specified hydrostatic test pressure (see 6.1) when applied in the water box or between the double tube sheets.

# 3.17.3 The design of the water boxes and double tube sheet assembly shall be based on 25,000 cycles to full design submergence and associated temperature variations and other transients. An analysis of fatigue loading on a cumulative damage basis shall be submitted to the agency concerned for approval with a copy to NAVSEC.

# 3.17.4 Where water boxes are not amenable to the fatigue loading analysis a proof test shall be conducted as specified in 4.3.4.

3.18 Condensers. Each condenser shall be a shell and tube design with water in the tubes and refrigerant in the shell. A combination condenser-chiller of integral construction is permitted.

# 3.18.1 The condensers shall be designed with removable water boxes or heads to permit tubes to be examined, cleaned or replaced as necessary. Suitable means shall be provided for venting entrained air from each inlet pass. Adequate condenser head drains shall be provided. Lifting lugs or eyes shall be provided to facilitate head removal.

# 3.18.1.1 Except on nuclear powered ships, condensers having copper-nickel alloy tubes, and tube sheets; and copper-nickel, cast bronze, nickel aluminum bronze, or nickel bronze heads need not be provided with zinc anodes or with the capability for installing zinc anodes. Provisions for installation of zinc anodes in accordance with MIL-A-19521 shall be made in all other sea water condensers. Where used the zincs shall be capable of being replaced without removing the heads.

3.18.2 Tube support plates shall be provided and installed so that the maximum unsupported tube length between supports or between a tube sheet and a support plate will not exceed 3 feet 0 inch.



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3.18.3 Baffles shall be installed as necessary to prevent direct impingement of high velocity vapor against the tubes to insure good distribution of the refrigerant to all parts of the condenser.

# 3.18.4 Tube sheets shall be machined and grooved to accommodate tubes expanded at the inlet and outlet ends. Holes for expanded tubes shall be drilled and reamed to a diameter of 0.751 inch with a tolerance of plus 0.005 inch, minus 0.000 inch for type II units and plus 0.007 inch, minus 0.000 inch for type I units. All holes at each face of the tube sheet shall be deburred and rounded off to a radius of 1/16 inch - 45 degree to remove the corner except that on the water inlet end of tubes on the outer (water) side of tube sheet the tube holes shall be belled and the tubes finished flush with the tube sheet.

3.18.5 Tubes shall be expanded into the tube sheet by means of an automatic tube expander controller conforming to MIL-C-17728.

# 3.18.6 Condenser heads shall be so proportioned as to provide sufficient area at all points for easy flow of the circulating water and for uniform distribution of water to all tubes in each pass.

# 3.18.6.1 Condenser head depth measured normal to the tube sheet shall be not less than one third of the inside diameter of the head measured parallel to the tube sheet.

# 3.18.7 The design of the condensers shall be such that the ordering length of tubes will be 96, 108, 120, 132, 144, 156 or 168 inches with a tolerance of minus 0+1/8 inch.

# 3.18.7.1 The ordering length includes the plain end section used to roll the tubes in the tube sheet.

# 3.19 Condensers for type I units. Sea water containing parts and components shall be designed to withstand, without damage, a hydrostatic pressure of 270 psi. Refrigerant containing parts and components shall be designed to withstand a pressure of 30 psi or 1-1/2 times the design working pressure, whichever is greater.

# 3.19.1 Condenser heads shall be constructed of cast gun metal, nickel-aluminum bronze in accordance with MIL-B-23921, valve bronze, nickel bronze or 90-10 copper-nickel alloy in accordance with MIL-C-15726. The water connections shall be of the nonferrous flanged type in accordance with MIL-F-20042. Bolting material including bolts, studs and nuts used to fasten water heads shall be nickel-copper in accordance with class B of QQ-N-281; aluminum bronze, stress relieved, in accordance with ASTM B-169; copper-silicon in accordance with ASTM B-98 or phosphor bronze in accordance with ASTM B-139.

# 3.19.2 Condensers for type I units shall have even number of passes if arrangements A, B or C are required and odd number of passes if arrangements D or E are required unless otherwise specified (see 6.1).

3.19.3 The tube sheets shall be constructed of 90-10 copper-nickel alloy conforming to MIL-C-15726. The tube sheet shall be not less than 1 inch thick and shall be welded to the condenser shell.

# 3.19.4 The condenser tubes shall be constructed of copper-nickel alloy in accordance with MIL-T-22214 for 90-10 composition. The tubes shall be 3/4 inch O.D. and shall be extruded fin type with minimum 0.049 inch wall thickness at the root of the fin portion. The tubes shall be installed in the tube sheet on a nominal triangular pitch of not less than 7/8 inch.

# 3.20 Condensers for type II units. Condensers for type II units shall be designed with an even number of sea water passes. Sea water containing parts and components shall be designed to withstand without damage a hydrostatic test of 1-1/2 times the sum of submergence pressure plus pump shut-off pressure as specified (see 6.1). Refrigerant containing parts shall be designed to withstand a pressure of 30 psi or 1-1/2 times the design working pressure, whichever is greater.

3.20.1 Condenser water heads shall be constructed of cast nickel-aluminum bronze, in accordance with MIL-B-23921, cast 70-30 copper-nickel alloy in accordance with MIL-C-20159 or 70-30 copper-nickel alloy in accordance with MIL-C-15726. Bolting material including bolts, studs and nuts used to fasten water head to the unit shall be K-monel in accordance with QQ-N-286.

3.20.1.1 Water heads shall be hemispherical or semi-ellipsoidal form. The semi-ellipsoidal form shall not be used when water nozzles and their reinforcements cannot be fully located within the central 40 percent of the water head diameter.



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# 3.20.1.2 Openings in the shell of the water heads for nozzles and the nozzles attached thereto shall be radial. Reinforcement of openings shall be integral with the water head shell or nozzle (or both). Separate pads or saddle type reinforcements shall be used. Welds shall be ground smooth and all corners and fillets shall be rounded to minimize notch effect.

3.20.1.3 Strengthening ribs shall not be used on water heads.

# 3.20.2 The condenser shall be of double tube sheet construction. A supplementary tube sheet shall be installed adjacent to each main tube sheet. This double tube sheet complex shall provide at least a 1/2 inch wide void space at each end of the condenser for detection of any seepage of sea water through expanded tube ends. This space shall be provided with vents and drains. The primary tube sheet shall be constructed of 70-30 copper-nickel alloy conforming to MIL-C-15726. The supplementary or inner tube sheet shall be constructed of 70-30 copper-nickel alloy conforming to MIL-C-15726 or steel. Steel inner tube sheets shall be in accordance with one of the following publications, except that impact tests of 3.20.2.1 shall be met: (a) ASTM A-105, grade I or II, (b) ASTM A-515, grade 55, 60, 65 or 70, firebox quality, (c) ASTM A-204, grade A or B or C, firebox quality and (d) ASTM A-266, class II. The supplementary tube sheet shall withstand the submergence test pressure required for the primary tube sheet. Thickness of each tube sheet shall be based on the sea water pressure specified, however, thickness shall be not less than 1 inch.

# 3.20.2.1 Impact test requirements. Steel material for submarine condenser inner tube sheets shall meet the special mechanical property requirements of Subarticle M-330 of Section III of the A.S.M.E. Boiler and Pressure Vessel Code. Impact values shall be met at a temperature of minus 30°F. or lower.

# 3.20.3 Condenser tubes shall be constructed of copper-nickel alloy in accordance with MIL-T-22214 for 70-30 composition. The tubes shall be 3/4 inch O.D. and shall be extruded fin type with minimum 0.065 inch wall thickness at the root of the fin portion. The tubes shall be installed in the tube sheet with a minimum ligament of 0.182 inch unless fatigue calculations for a submarine condenser establish the need for a thicker ligament, in which case such design value shall become the minimum ligament thickness. Out-of-roundness of these holes shall not exceed 0.001 inch.

# 3.20.3.1 Integral finned condenser tubes for type II units shall be inspected after finning by the eddy current method. The sensitivity of the device used for eddy current testing shall be adjusted to detect defects (splits, overlaps, circumferential grooves, etc., caused by the finning operation) in the finned section that extends into the root wall 0.004 inches or greater. Tubes having defects in excess of this depth shall not be used on type II units.

# 3.20.4 For submarine condensers, the following items are designated as critical materials. Each of these items shall have material traceability including heat lots, chemical and physical test reports. Critical materials shall be designated on drawings and shall consist of the following:

(a) Tube sheets (both inner and outer)
(b) All sea water vent and drain connections including double tube sheet void space vents and drains
(c) Tubes
(d) Waterboxes including all parts, subparts and access covers
(e) Nuts, studs and bolts associated with sea water containing parts

# 3.20.4.1 Traceability as indicated in 3.20.4 means the ability to use the material identification marks to determine which records are applicable.

# 3.20.4.2 The material identification marking shall indicate the material, category of material (specification, grade, composition, etc.), lot or serial numbers and manufacturers symbol. Where space is not available the manufacturer may use a code to provide traceability.

# 3.20.4.2.1 The following shall be marked:

(a) Materials
(b) Components when they are ordered separately (see 3.20.4.2.2(a)).



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# 3.20.4.2.2 Where to mark:

(a) Marking is required on only one place on a completed assembly, e.g., on valve body only, with no marking on other valve components, (see 3.20.4.2.1(b)).
(b) Marking shall be accessible to permit identification without disassembly, except for justifiable situations, when alternate methods of identification shall be used to identify these materials.
(c) Marking shall be located where it is not detrimental to the end use of the item.

# 3.20.4.2.3 How to mark:

(a) Marking shall be legible.
(b) Marking shall be permanent whenever the type, size and condition of material permits, except as stated in (c).
(c) Permanent marking is not required for small items or for materials whose type and condition precludes the use of permanent marking. However, such items shall be marked as follows:

(1) Package small items and label the package with the required marking.
(2) Small items shall be repackaged with the required marking on an envelope, tag, label, or tape when issuing small quantities from a defined lot.

# 3.20.4.2.4 Marking tools and methods:

(a) Permanent marking shall be applied by a method which will not reduce thickness below minimum design requirements, induce detrimental stresses, contaminate or cause maloperation.
(b) Unless specifically approved, marking with impact tools is prohibited except that round bottom tools may be used for marking on a raised pad, flange edge, fastened head or end, or other low stress area providing that the impression does not exceed 0.010 inch or reduce wall thickness below design requirements.
(c) Unless specifically approved, electric arc etching is prohibited.
(d) Use of a vibrating pencil or electro-chemical etching is permitted.
(e) Roll marking is permitted on bar stock where the die imprint shall not occur on a critically stressed surface.

3.21 Water chillers. Each water chiller shall be of the shell and tube flooded type with water flowing through the tubes and refrigerant in the shell. The chiller shell shall be of adequate diameter or provided with eliminators to avoid liquid entrainment back to the compressor.

3.21.1 Chiller tubes shall be installed with nominal triangular pitch of 7/8 inch. The tubes shall be 3/4 inch O.D. copper of the integral fin type with a minimum wall thickness of 0.049 inches at the root of the fin section of the tube. The tubes shall comply with MIL-T-22214 except the material shall be in accordance with WW-T-799. The chiller tubes shall be in accordance with 3.18.7 and 3.18.7.1.

3.21.2 The tube sheets, heads and other water containing parts shall be nonferrous. Tube sheets shall have a minimum thickness of 7/8 inch. Water connections shall be of the non-ferrous flanged type in accordance with MIL-F-20042.

3.21.3 Provision shall be made for purging air and draining water from the chiller heads. Bolts and nuts for fastening the heads shall be aluminum bronze, phosphor-bronze or copper-silicon alloy. All joints on the chiller shall be silver brazed or welded. The chiller shall be insulated in accordance with MIL-STD-769.

# 3.21.3.1 Chillers shall have even numbers of water passes unless otherwise specified (see 6.1).

3.21.3.2 The chiller head inlet water nozzle shall be located at the opposite end of the unit from the condenser sea water inlet nozzle unless otherwise specified (see 6.1).

# 3.21.4 A sight glass or bulls-eye shall be provided to indicate the refrigerant level in the chiller during shut down.



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3.22 Interstage coolers and liquid subcoolers.

3.22.1 Interstage coolers. If necessary for efficient operation, an interstage cooler shall be provided for each unit to precool the condensed liquid to a temperature corresponding to the interstage pressure. The flash gas should be adequately separated to prevent liquid entrainment to the interstage impeller and decrease the amount of flash gas through the water chiller.

3.22.2 Liquid subcoolers. If necessary for efficient operation, a liquid subcooler shall be provided for each unit to precool the condensed liquid to a temperature approaching the entering sea water and decreasing the amount of flash gas through the water chiller.

# 3.23 Structural base. The base shall be fabricated of steel with welded joints designed to withstand shock and to support the weight of the complete assembly. Sufficient rigidity shall be provided to insure that proper alignment of the assembled equipment can be maintained in service. Positioning of components to bedplate shall be maintained by use of dowel pins of a size required for the application. Where type II unit is specified, the assembled unit shall be arranged structurally for mounting, using resilient mounts in a horizontal plane passing approximately through the center of gravity. Resilient mounts of a type specified in MIL-STD-740 shall be provided by the contractor for type II units.

# 3.24 Controls. Primary capacity control shall be obtained by means of an automatic capacity reducing device to regulate capacity down to approximately 15 percent of design load at constant speed by varying the volume of gas handled by the compressor. A suitable operator shall be provided to automatically position the prerotation vanes in response to a temperature control system sensing the leaving chilled water temperature thereby maintaining an essentially constant chilled water leaving temperature. The control shall be designed to operate satisfactory with an ambient temperature of 50°C. The control system shall be factory wired, pipe connections made and include at least the following:

(a) Chilled water temperature control with operating point adjustment.
(b) Compressor start-stop push button and operating signal light.
(c) A manual operator to control the prerotation vanes.
(d) A compressor motor current limiting device.
(e) Protection against short cycling with an anti-recycling timer with a lockout of approximately 15 minutes. An emergency override shall be provided for this control.
(f) Inter-locking control as required by 3.16.3.1 and 3.16.3.2.
(g) Ammeter.
(h) Inter-lock to activate crankcase heaters and oil heater temperature control when unit is not in operation.
(i) Safety controls as follows:

(1) High oil temperature.
(2) High refrigerant pressure.
(3) Low refrigerant temperature.
(4) Low oil pressure.
(5) Condenser water failure (type I units only).
(6) Differential chilled water pressure failure.
(7) Chilled water low temperature.
(8) Compressor motor high temperature cut out (class 1 units only).

A visual signal shall be provided for each safety control to indicate which condition has caused stopping of the motor. An audible alarm shall be provided to indicate when plant shuts down for any reason other than power failure. A manual cut-out shall be provided for the audible alarm.

3.24.1 Semiconductor devices shall conform to MIL-S-19500 and shall be a type listed in MIL-STD-701 or be approved for the application by NAVSEC or the agency concerned.

# 3.24.2 A control panel shall be provided with refrigerant suction and discharge pressure gages, chilled water inlet gage, oil pressure gage, chilled water inlet and outlet thermometers, condenser water inlet and outlet thermometers, refrigerant suction and discharge thermometers, and oil thermometers. The control panel shall be so designed to permit accessibility for repair and maintenance with adequate safety protection. The enclosure shall be compartmented to separate fluid and electrical devices whenever possible. Provisions shall be made to drain off fluid leakage.



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# 3.24.2.1 The pressure indicators shall be in accordance with MIL-I-18997. The pressure indicators shall be 3-1/2 or 4-1/2 inch dial and interchangeable in Navy Standard mounting. The following pressure indicators shall be provided for remote installation:

(a) Refrigerant suction with range 30 inches vacuum to 15 pounds.
(b) Refrigerant discharge with range 30 inches vacuum to 30 pounds.
(c) Chilled water inlet with range 0 to 100 pounds.
(d) Sea water condenser inlet with range 0 to 200 pounds (for type I).
(e) Oil pressure with range 0 to 200 pounds.
(f) Air pressure, if pneumatic controls provided.

# 3.24.2.1.1 The sensing line for each pressure indicator shall contain valve(s) and a calibration connection arranged so that the pressure indicator can be isolated for repair and checked for accuracy using an external pressure source without removing the pressure indicator from the control panel.

# 3.24.2.2 The thermometers shall be in accordance with MIL-T-19646 except they shall have 3-1/2 or 4-1/2 inch dial size and the bulb size may be as small as 3/8 inch.

(a) Chilled water inlet and outlet - range minus 40° to 180°F.
(b) Sea water condenser inlet and outlet - range of minus 40° to 180°F.
(c) Compressor discharge - range of 20° to 240°F.
(d) Compressor suction - range of minus 40° to 180°F.
(e) Oil sump-range of 20° to 240°F.
(f) Oil temperature from bearings - range of 20° to 240°F.

# 3.24.2.2.1 Thermometer wells for sea water application shall be in accordance with MIL-W-24270 and shall be of 90-10 copper-nickel (type I units), 70-30 copper-nickel (type II units), or nickel copper alloy.

# 3.24.3 Solenoid valves where provided shall be in accordance with the requirements of MIL-R-16743 as applicable.

# 3.24.3.1 Solenoid valves for water service shall be provided with coils of the encapsulated type with integral terminals. Solenoid enclosures shall be water tight.

# 3.25 Dehydrator. Dehydrators or filter dehydrators where provided shall be in accordance with the requirements of MIL-R-16743 as applicable and shall have a moisture indicator or sight glass installed to indicate when cartridge replacement is required.

# 3.25.1 A dehydrator shall be provided to remove moisture and foreign particles from the refrigerant within the unit. It shall be piped and valved such that refrigerant can be circulated through the dehydrator while the unit is operating and the cartridges replaced without stopping the unit.

# 3.26 Water regulating valves - (type I units only). A temperature actuated water regulating valve shall be provided with each unit for installation in the condenser sea water discharge piping.

# 3.26.1 The water regulating valve shall be a modulating type which will maintain a relatively constant condenser sea water discharge temperature by modulating the flow of water required for the condenser. It shall be of the direct-acting or pilot controlled type pneumatically operated utilizing a thermostatically controlled 20 psig air supply. The valve shall have a nominal pressure rating of 250 psig. Flow through the valve shall be from under the seat tending to open the valve. The valve shall be designed or adjusted such that water will flow through the unit even when the valve is fully closed.

# 3.26.2 The valve shall be constructed of non-ferrous or corrosion-resisting material suitable for sea water. The valve body shall be made of valve bronze conforming to ASTM B-61. In addition the internal metal parts subject to corrosion or erosion shall be made of nickel-copper. The valve shall have flanged connectors in accordance with MIL-F-20042. The valve connections shall be the same size as the condenser discharge connection or a transition piece shall be provided.

# 3.26.3 The valve shall be adjustable and the actuator shall be sized to fully stroke the valve with a maximum control signal of 15 psig, assuming a maximum pressure differential across the valve plug of 60 psig. The valve shall be sized to pass the maximum design water flow with a 15 psi drop through the valve.



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# 3.26.4 An adjustable pneumatic thermostat with remote bulb located in the condenser sea water discharge line shall be provided to control valve operation. The pneumatic controls shall provide full signal range to the water regulating valve actuator over a range of 70° to 100°F.

# 3.26.5 The valve, actuator, and thermostatic controls shall be capable of passing the HI shock requirements (see 3.13).

3.27 Purge unit. The purge unit shall include the necessary controls, piping and valves to isolate the purge system from the main refrigerant system. It shall function automatically to remove air, water vapor and the other non-condensables from the system and shall be capable of manual operation. The purge unit shall also condense, separate and return to the system any refrigerant present and shall have means of visually indicating when water is present. The system shall consist essentially of a small refrigerant compressor unit or pump, a purge shell and heat exchangers as necessary, adequate safety devices, means of containing oil in the purge system if used, pressure gages and all necessary piping, valves and controls.

# 3.27.1 Except for the compressor, all parts of the purge circuit shall be of non-ferrous material or corrosion-resistant chrome-nickel-steel and discharge valves for the compressor shall be of non-ferrous construction.

3.27.2 Where the purge unit is mounted remote from the type II refrigerating unit on submarine application, it shall be provided with resilient mounts of a type specified in MIL-STD-740.

# 3.27.3 Purge unit motor. The purge unit motor shall be in accordance with the requirements of 3.16.4 and 3.16.4.1.

# 3.27.3.1 Purge unit motor controllers when required shall be across-the-line, magnetic, with LVP (non-automatic) type, in accordance with MIL-C-2212.

# 3.27.4 For type II units the purge compressor and purge system shall be designed to operate satisfactorily and be capable of discharging non-condensables to the foul gas storage tank at the foul gas storage tank pressure.

# 3.28 Vacuum pump. A vacuum pump shall be provided with the unit to evacuate the refrigerant envelope of the unit after repairs have been completed to effect dehydration. The vacuum pump shall be capable of evacuating the refrigerant envelope of the unit to a vacuum of 29.72 inches of mercury in 2 hours or less. The evacuating time is predicated on the unit containing no free moisture and a standard atmospheric pressure of 29.92 inches of mercury. A 6 to 8 inch dial absolute pressure gage with at least the following range and accuracy shall be provided:

| | | |
|---|---|---|
| Range | - | 0 - 30 inches in Hg absolute |
| Accuracy | - | 0.2 percent of full scale range |
| Sensitivity | - | 0.05 percent of full scale range |

# 3.28.1 Vacuum pump motor. The vacuum pump motor shall be in accordance with the requirements of 3.16.4 and 3.16.4.1.

# 3.28.2.2 The vacuum pump controller shall be across-the-line, magnetic, with LVP (non-automatic) type, in accordance with MIL-C-2212.

# 3.28.3 The vacuum pump, less vacuum gage, shall be capable of passing the high impact shock tests specified in MIL-S-901 for grade B equipment.

# 3.28.4 The vacuum pressure gage may be stored separate from the vacuum pump.

# 3.28.5 Suitable vacuum hoses shall be provided to connect the vacuum pump to the air conditioning units. The hoses shall have adequate size, length, and correct end connection.

# 3.29 Refrigerant transfer. The centrifugal unit and purge unit for type II installations shall contain valves and connections and be arranged such that the refrigerant envelope of the unit and an external refrigerant storage tank or drum(s) can be pressurized using the purge unit to facilitate refrigerant transfer.

# 3.30 For the type II unit, it is essential that every precaution be taken to minimize the contamination of the submarine atmosphere with refrigerant vapors during operation, maintenance and repair.



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# 3.30.1 The unit shall include valved connections to drain the liquid refrigerant charge to a closed container. Similarly, the oil system shall also contain valved connections for draining the lubricating oil to another closed container. A portable closed container(s) with suitable valved connections shall be provided for draining the oil from the units. The volume of the container(s) shall be large enough to hold the complete oil charge from one unit. Means shall be provided to pump out the refrigerant envelope of the system so that the residual refrigerant can be condensed and returned to the receivers prior to opening or disassembling the unit. If water is required to condense the residual refrigerant, the condenser of the pump out equipment shall be designed to utilize chilled water at the indicated pressure (see 3.1.1) and temperatures (see 6.1).

# 3.30.2 The pump out equipment shall be capable of pumping out the refrigerant envelope of the unit to a vacuum of 27.5 inches of mercury in 3 hours or less. In addition, means shall also be provided to evacuate the refrigerant envelope of the unit after repairs have been completed to effect dehydration. The evacuating equipment shall be capable of evacuating the refrigerant envelope of the unit to a vacuum of 29.72 inches of mercury in 2 hours or less. The evacuating time is predicated on the unit containing no free moisture and a standard atmosphere pressure of 29.92 inches.

# 3.31 Onboard repair parts. The procedure for furnishing and identifying onboard repair parts shall be in accordance with MIL-P-15137. The number of sets shall be as specified (see 6.1). The onboard repair parts shall include those specified in table I, as applicable, and shall constitute one set. The quantities specified refer to the number required for each quantity of similar equipment furnished on ship. If the contractor furnishes equipment or components identical to equipment or components previously furnished to the Government for which an Allowance Parts List exists, then the onboard repair parts shall be in accordance with the existing Allowance Parts List.

# Table I - Onboard repair parts.

| Item | For Determination of Onboard Allowed Quantity Per Ship<br>Multiply the allowance factor in the applicable number of components installed per ship column by the quantity in one unit. Decimal quantities are rounded to the nearest whole number. In the event any quantity computes to less than one; i.e. .1, .0, a quantity of one is used. In no event will a quantity greater than 999 be used. | | | | | |
|---|---|---|---|---|---|---|
| | Factor Code | No. of components | | | | |
| | | 1 | 2 | 3 | 4 | 5-8 |
| Compressor repair parts | | | | | | |
| Complete set of seal assemblies of each type and size. | D | 1 | 2 | 3 | 3 | 4 |
| Complete set of bearing assemblies of each type and size. | D | 1 | 2 | 3 | 3 | 4 |
| Complete sets of gaskets. | D | 1 | 2 | 3 | 3 | 4 |
| Oil pump | D | 0.33 | 0.66 | 1 | 1 | 2 |
| Set of bushings of each type and size. | P | .33 | .66 | 1 | 1 | 2 |
| Oil filter | D | 1 | 2 | 3 | 3 | 4 |
| Chiller repair parts | | | | | | |
| Set of gaskets of each type and size. | D | 1 | 2 | 3 | 3 | 4 |
| Float valve assembly including ball. | P | 0.33 | 0.66 | 1 | 1 | 2 |
| Condenser repair parts | | | | | | |
| Sets of tube sheet or water head gaskets | D | 1 | 2 | 3 | 3 | 4 |
| Purge and recover unit-where provided | | | | | | |
| Compressor, or pump, as applicable. | Q | 0.25 | 0.50 | 0.75 | 0.75 | 1 |
| V-belts set. | D | 1 | 2 | 3 | 3 | 4 |



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Table I - Onboard repair parts - (cont'd)

| Item | For Determination of Onboard Allowed Quantity Per Ship<br>Multiply the allowance factor in the applicable number of components installed per ship column by the quantity in one unit. Decimal quantities are rounded to the nearest whole number. In the event any quantity computes to less than one; i.e., .1, .0, a quantity of one is used. In no event will a quantity greater than 999 be used. | | | | | |
|---|---|---|---|---|---|---|
| | Factor Code | No. of components | | | | |
| | | 1 | 2 | 3 | 4 | 5-8 |
| Miscellaneous - where provided | | | | | | |
| Heater element. | P | 0.33 | 0.66 | 1 | 1 | 2 |
| Special tools, thermometers, gages, valves, solenoid valve coils and controls of each type and size. | P | 0.33 | 0.66 | 1 | 1 | 2 |
| Pre-rotation vane positioner. | P | 0.33 | 0.66 | 1 | 1 | 2 |
| Oil pump and motor assembly. | F | 1 | 1 | 1 | 2 | 2 |
| Dryer cartridges | C | 2 | 3 | 4 | 4 | 5 |
| Motors and controllers<br>See MIL-M-17059, MIL-M-17060 and MIL-C-2212 as applicable. | | | | | | |
| Refrigerant and oil | | | | | | |

1. Replacement lubricating oil charge for each unit in 1 gallon containers.
2. The replacement refrigerant charge for type I units shall be supplied in 100 pound drums and for type II units in 50 pound cylinders. The quantity of replacement refrigerant supplied shall be in accordance with the following:

| Number of units installed onboard ship | Quantity of replacement refrigerant supplied |
|---|---|
| 1 to 3 units | 125 percent of a complete charge[1/] |
| 4 units | 225 percent of a complete charge |
| 5 to 8 units | 325 percent of a complete charge |
| 9 units and above | 425 percent of a complete charge |

[1/] Complete Charge=Refrigerant Charge for 1 unit i.e., if a ship has 5 units installed, each requiring 560 pounds of refrigerant, the quantity of replacement refrigerant supplied shall be - 560 x 3.25 = 1820#

| Type I | Type II |
|---|---|
| 1820/100 = 19 (100# drums) | 1820/50 = 37 (50# cyclinders) |

3.32 Identification plates. Identification plates shall be made of brass (type A), aluminum or corrosion-resisting steel (type H) in accordance with MIL-P-15024. Information plates shall be made of laminated plastic, (type F), aluminum or corrosion-resisting steel (type H) in accordance with MIL-P-15024.

3.33 Painting. For equipment operating at a temperature of 300°F or less, all external unmachined surfaces of metal parts, except corrosion-resisting steel, copper or brass shall be thoroughly cleaned, and coated with paint in accordance with the following specifications or equivalent materials:

(a) One coat of pretreatment coating MIL-C-15328.
(b) One coat of zinc chromate primer TT-P-645 or one coat of red lead primer MIL-P-17545.
(c) One finish coat of light gray equipment enamel MIL-E-15090.



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# 3.34 Drawings. Drawings delineating the equipment shall be furnished. Drawings shall be type II, III, or IV in accordance with MIL-D-1000/2. The drawings shall be for intended use categories A, B, G and H as delineated in accordance with MIL-D-1000. Separate drawings need not be furnished for each category. Parts lists, bill of material lists, etc., shall be included on equipment drawings. Certification data drawings should include diagrams and drawing list.

# 3.34.1 Assembly drawings. Assembly drawings, type II or IV shall be provided for each of the following:

(a) Centrifugal refrigerating unit
(b) Compressor
(c) Condenser
(d) Chiller
(e) Speed increasing gear assembly (separate casing)
(f) Coupling
(g) Purge unit
(h) Controls
(i) Instruments
(j) Ancillary equipment or components

# 3.34.1.1 The assembly drawing shall show outline, mounting, attachment and connection dimensions including methods and sizes of fastenings and clearances for installation and servicing plus supplementary data necessary to permit shipyard installation without supplier's assistance. The drawing shall illustrate design, construction, operation, (or function), identity of parts, model number, total wet and dry weight and center of gravity of the component or unit. All connections requiring interface with shipbuilder installed piping shall be delineated as to size, type and three plane dimensional locations indicated with a tolerance no greater than plus or minus 1/4 inch. Performance data and curves shall be included with design conditions. Curves shall be provided to show how the capacity of the unit and power required change with varying sea water temperatures when the leaving chilled water temperature is set at the design point (see 6.1) and about 3° to 5°F. above the design point. In addition, a curve shall be provided to show how the power required varies with capacity when the condenser is at design conditions. The radii of gyration of the complete wet unit about each of the three principal axes should be shown for the centrifugal refrigerating unit. The drawing shall be sufficiently complete to show compliance with equipment specification requirements. Where necessary to illustrate compliance, details may be included on the assembly drawing or provided as separate drawings. Sub-assembly drawings shall be furnished where assembly drawings do not adequately describe and identify sub-assembly parts and components.

# 3.34.2 Certification data, type III drawing(s) shall be provided. The drawing(s) shall include a reference drawing list of all the refrigerating unit assemblies, components, parts, control system items and ancillary components being furnished on the contract or order. Diagrams shall also be provided. The diagrams shall include a refrigerant and water piping diagram, pneumatic or electrical control diagram, lubricating oil and electrical wiring diagram. The diagrams shall show by symbolic representation all fluid piping, internal passages, pipe sizes, electrical interconnections, components, accessories, control and associated instruments, performance characteristics as applicable, direction of flow, operating sequence and connections required by others for operation. For example, a piping diagram will include associated units, controls, gages, thermometers, valves, piping accessories, sea water and chilled water connections, as applicable. An electrical wiring diagram will include and show circuits between motor controllers, motors, controls, switches and accessories as applicable for an understanding of operating sequence. All symbols used for equipment shall be given a piece number and identified in the list of materials with the following information:

(a) Piece number
(b) Quantity required
(c) Descriptive name
(d) Manufacturer's name
(e) Manufacturer's model or identifying number
(f) Manufacturer's drawing number
(g) Weight
(h) Characteristics



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# 3.34.2.1 The characteristics shall include ordering information necessary to specifically qualify or supplement data described in reference drawing or manufacturer's model and part number. A separate table shall be used for the various components where necessary and shall include the following:

(a) Square feet of tube surface, number of passes, types of surface (finned or bare), design fluid flow (gpm), fluid velocity, pressure drop, entering and leaving temperatures and heat transfer, as applicable.
(b) Complete data for the assembly of equipment shall be provided including the model number, capacity, design capacity rating conditions, electrical requirements, test pressure, specification acceptance tests conducted (identify each test), approved by, date approved and contract or order number.

# 3.34.3 Detail drawings shall be furnished of all parts and sub-assemblies necessary for evaluation of the equipment, and all parts necessary for maintenance and overhaul of the equipment. Drawings shall show all essential fabrication details including welding requirements and symbols. Sub-assemblies whose parts cannot be procured or serviced individually should be shown as a single part. Multi-detail drawings are preferred, but mono-detail drawings may be used. Drawings are not required for those parts which are in common commercial use and can be referenced to commercial standards.

# 3.34.4 Electrical component drawings. In addition to drawing requirements specified herein, the content and format of electrical component drawings shall conform to the requirements of the applicable component specification.

# 3.34.5 Requirements for tabulating index cards and microfilm of engineering drawings mounted in aperture cards.

# 3.34.5.1 Microfilm mounted in aperture cards shall be supplied with coverage of all drawings applicable to all items of equipment. Microfilm shall be type I, class 1 or type II, class 2 in accordance with MIL-M-9868 one sheet per frame format shall apply. A set of cards shall consist of that number of cards necessary for the required coverage.

# 3.34.5.2 Tabulating index cards shall be supplied for aperture cards. A set of tabulating index cards shall consist of an index card for each aperture card in a set of microfilm aperture cards as defined herein.

# 3.34.5.3 Selection and preparation of aperture and tabulating index cards shall be in accordance with MIL-STD-804, MIL-M-38761, and MIL-C-9877.

# 3.34.5.4 Dual purpose engineering document card DD form 1562. Card code "T", lower legends of MIL-STD-804 shall be obtained commercially and used in the preparation of all aperture and tabulating index cards.

# 3.34.5.4.1 Aperture and tabulating index cards shall be punched and interpreted in accordance with MIL-STD-804 and the following:

(a) Card columns 31-32; punch the frame number of a multiframe drawing. A single frame drawing shall have frame number "1" entered in right hand column. This information shall be interpreted using interpreter bar locations 29U30.
(b) Card columns 33-34; punch the number of frames required to completely photograph the drawing. The number of frames digit shall be entered in the right hand column. This information shall be interpreted using interpreter bar locations 31U32.
(c) Card columns 35-36 shall be left blank.
(d) Card columns 48-49; insert control activity symbol "HR". This information shall be interpreted using interpreter bar locations 51U52.
(e) Card columns 53-80; punch the drawing title in the tabulating index card only. This information shall be interpreted on both index and aperture cards using interpreter bar locations 12L39.

# 3.34.5.5 Security requirements. Aperture cards containing classified drawings shall be identified and marked in accordance with MIL-STD-804.

# 3.34.5.6 Responsibility to update sets of microfilm aperture cards and tabulating index cards supplied to the Government. Microfilm mounted in aperture cards and tabulating index cards shall be supplied for drawings generated or revised during the time period between initial submittal of card sets and expiration of the contractors responsibility for last ship set of equipment.



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3.34.6 Model development drawings. Drawings and other design information from which the planning yard can develop a full scale mock-up of the air conditioning unit(s) being supplied shall be provided when required (see 6.1). The mock-up will be used to develop the space arrangement and piping layouts in the area of the air conditioning unit(s). The drawings supplied to the planning yard shall contain all of the information required by the planning yard to develop the mock-up and at least the following information:

(a) Dimensional locations and size of components and connections
(b) Mounting arrangement and locations
(c) Access areas and clearance required for service and maintenance
(d) Contours of components

3.35 Manuals. Manuals furnished shall comply with type III of MIL-M-15071, photo views of the equipment shall be included as a part of the general description. A section shall be provided containing reduced copies of all drawings required to amplify or illustrate the text including diagram and assembly drawings. The quantity of manuals furnished shall be as specified (see 6.1). Where identical manuals have been distributed to addressees, additional manuals need not be furnished except two manuals packaged with the equipment and the quantity of manuals specified for Naval stock.

# 3.35.1 The contractor shall validate or have validated all manuscripts for new and revised manuals, including changes, furnished under the terms of the contract. In those cases where validation is performed at a subcontractor's plant, the prime contractor is responsible for the quality of the manual manuscript and must provide verifiable evidence of the adequacy of the validation performed. Validation must be completed and all corrections made and validated prior to presentation of any part of the manuscript to the Government for verification. Validation will include but not be limited to the following:

(a) All written information, engineering drawings, and art work in the manual must be compared to the related physical equipment to assure that they actually delineate that equipment.
(b) A demonstration, by actual performance, of all instruction and procedures in the manual on the physical equipment shall be accomplished. This demonstration may coincide with the maintainability demonstration (see 4.4.5). Items of the following nature need not be performed.

(1) Destruction of material such as disassembly of electronic components or removal of condenser tubes, is not required.
(2) Boring, grinding, and other shaping repair procedures need not be actually executed.

3.35.1.1 At the discretion of the Government some of the procedures and instructions in the manual need not actually be performed where it is evident through visual examination or by following a logical sequence of steps that the procedures or instruction can be performed as indicated.

# 3.35.2 The procuring activity will verify all manuscripts or parts of manuscripts for new and revised manuals, including changes. All discrepancies will be brought to the attention of the contractor. All corrections must be made and verified. Verification will include demonstration by actual performance of instructions and procedures within the manual of the physical equipment using a valid sampling plan as a minimum. Verification is to be performed simultaneously with validation whenever possible.

# 3.35.3 Content assurance pages. After manuscript validations and verification have been completed, content assurance pages are to be inserted into each manual in its final form and into the corrected, accepted reproducible copy. These additional pages are to be inserted immediately after the title page. Cognizant activities and contractors are responsible to provide information for their respective sections of these pages. Contractors are responsible for the insertion of these pages. The following information, as a minimum, is required on these additional pages:

(a) Validation page
(1) Contractor's full identification (if other than prime, full identification of both must be indicated.)
(2) Contract number(s).
(3) Chapters and sections validated and the date each was accomplished.
(4) Chapters and sections not validated. (Justification for noncompliance must be delineated in separate correspondence).
(5) Name, signature, and authority of validating officer.
(b) Verification page
(1) Cognizant Government activity.
(2) Chapters and sections verified and the date each was accomplished.



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(3) Chapters and sections not verified.
(4) Name, signature, and authority of verifying officer.

# 3.35.4 Contractors are required to validate or have validated one of each type of existing manual(s) required to be supplied by them under the terms of the contract. If there is a content assurance page within the manual that indicates a complete validation, or any part thereof, performance of validation of the manual or part need not be repeated.

# 3.35.5 Cognizant activities will verify one of each type of existing manual(s) required to be supplied under the terms of the contract using a valid sampling plan as a minimum. Discrepancies in manuals that accompany Government supplied equipments or systems will be brought to the attention of NAVSEC. If there is a content assurance page within the manual that indicates a complete verification, or any part thereof, performance of verification of the manual or part need not be repeated.

3.36 Workmanship. Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality inspection system. The supplier shall provide and maintain an inspection system acceptable to the Government. The system shall be in accordance with MIL-I-45208. The supplier is in no way relieved of the final responsibility to furnish refrigerating units meeting the requirements of this specification.

4.2 Tests. Quality conformance and production tests shall be conducted. Instruments containing mercury shall not be used at any time.

# 4.3 Quality conformance tests. Quality conformance tests include shockproof, and capacity tests specified in 4.3.3 through 4.3.5. When equipment is furnished which is of the same design, type, and size as equipment previously furnished which has passed the quality conformance tests, these tests need not be performed, provided the supplier furnishes actual test, examination or other verifiable quality data acceptable to NAVSEC.

4.3.1 Where a manufacturer is unable to conduct these tests at his own plant, he may arrange to have them conducted at a commercial laboratory or Government laboratory suitably equipped to conduct such tests. Where shock tests are conducted at a Government laboratory, copies of the applicable approved drawings shall accompany the units.

4.3.2 Equipment previously shock tested and accepted need not be retested where evidence can be submitted to the command or agency concerned that the original item was subjectd to shock tests and successfully passed, and that the design of the proposed equipment, its intended service and shipboard installation are such as to result in an equal or better degree of shock resistance.

# 4.3.3 Shock test. Unless otherwise specified (see 6.1), one complete unit of each type, design and size shall be subjected to the high impact shock tests specified in MIL-S-901 for grade A equipment. Unless otherwise specified (see 6.1), class III of MIL-S-901 shall be used. The unit shall be tested in an operating condition (see 3.13.1). The design of the foundation used to shock test the unit or units shall be provided by the contractor unless otherwise specified (see 6.1). The foundation design shall provide adequate service openings where required. The foundation required for shock testing shall be provided by the contractor.

4.3.3.1 Equipment shall be considered to have failed to pass the shock tests in the event of any of the following:

(a) Breakage of any parts, including mounting bolts.
(b) Appreciable distortion or dislocation of any part or otherwise presenting a hazard to personnel or other equipment.
(c) Significant effect on performance.
(d) Leakage of refrigerant.



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4.3.3.2 Equipment which has been subjected to the high-impact shock test and has failed to conform to the requirements will not be acceptable.

# 4.3.3.3 Equipment which has been subjected to the high-impact shock test and has successfully passed this test shall be considered acceptable for such service as the command or agency concerned may authorize. Such equipment shall be reconditioned by the manufacturer as follows:

(a) Minor deformations including alignment.
(b) All bearings shall be replaced.
(c) Each part shall be carefully examined by the manufacturer and any part which he considers substandard shall be replaced.
(d) The units shall be cleaned, leak tested and dehydrated in accordance with the requirements specified herein.

Minor deformations shall be defined as those which do not cause unqualified rejection of the design under high-impact shock test but which are in excess of the design dimensional tolerances specified on the applicable drawings.

# 4.3.4 Proof test. For submarine condensers which will be subjected to full submergence pressure, the structural adequacy of the inlet-outlet waterbox shall be verified by a proof test if the configuration or other factors of the design are such that proof testing is required by SDB-63. If required, the proof test shall consist of a qualitative survey of the waterbox, utilizing brittle coating (stress coat or its equivalent), or photostress (molded birefringent coatings) techniques as deemed suitable, in order to determine the location of maximum strain. Quantitative measurement of the magnitude of maximum principal strain in the waterbox shall be obtained from strain gages of 1/8 inch gage length or smaller. Waterbox loadings shall be comprised of a combination of design submergence pressure in conjunction with piping reaction loadings applied to the nozzles. The nozzle loads shall be calculated as specified in 3.17. A suitable loading jig incorporating mechanical or hydraulic jacks shall be provided to induce external loads on the nozzle flanges. The following combinations of loads shall be applied:

(a) Internal pressure and external loads applied to one flange.
(b) Internal pressure and equal external loads applied to both flanges.

# 4.3.4.1 Since the angular orientations of the applied bending moments to give the maximum stress for each load combination are unknown, and the location of maximum stress is also unknown, the angular orientations of the applied bending moments for load combinations (a) and (b) of 4.3.4 shall correspond to increments of 45 degrees around the nozzles from 0 to 360 degrees in the first run, and a minimum of 9 degree increments shall be used in the 45 degrees maximum stress sector in the second run. The location, magnitude and direction of the maximum principal strain (stress) can be determined by brittle coating or photostress, but must be checked with electric resistance strain gages. If the maximum value of principal strain (stress) thus determined in any location of the waterbox (by either load combination (a) or (b)) of 4.4.4 does not exceed the allowable strain (stress) range for a 25,000 cycle life as determined by the Modified Goodman Diagram and Fatigue Curve for the particular material involved, the waterbox design is acceptable. The contractor shall submit to NAVSEC and the agency concerned an outline of the test procedure to be used, describing test fixtures, proposed instrumentation and procedures involved. The Government inspector shall monitor the set-up for test, procedures, and taking of data. Certification that a proof test in accordance with the above has been previously performed on equipment identical to that being furnished hereunder will suffice for the above requirements.

# 4.3.5 Capacity test for type I unit. A capacity test shall be conducted on one type I unit of each design, type and size to determine if the specified capacity is reached. The unit shall be adjusted to deliver the specified chilled water temperature and chilled water flow and operated with at least 88°F. condensing water and nominal 440 volt, 3 phase, 60 cycle power. The test also shall determine the ability of the machine to operate automatically at approximately 15, 50 and 100 percent of capacity with leaving water temperature and flow constant at design conditions, and condenser water entering temperature and flow rate constant at design conditions, when operating with controls. When testing the unit at 15 and 50 percent of rated capacity the chilled water flow may be reduce to 50 percent of design conditions. Instrumentation shall have an accuracy of at least plus or minus 5 percent. Chilled water temperature shall be measured to within 1/5 of a degree F. The capacity test shall be run using chilled water flows of both 3.6 and 4.5 gpm per ton (approximate) (see 3.2.1).



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4.4 Production tests. Production tests shall include the following:

(a) Noise and vibration (type II unit).
(b) Pressure and leak.
(c) Capacity (type II unit).
(d) Operation.

4.4.1 Noise and vibration for type II unit. Airborne and structureborne noise level tests, in accordance with MIL-STD-740, shall be made on each unit. Components shall be noise tested in normal horizontal position at approximately 15, 50 and 100 percent of rated capacity. The maximum allowable noise levels shall be as specified (see 6.1).

4.4.1.1 For nonrotating, nonreciprocating machinery, if the first two units of a design are below the required limits by 10 db or more at all frequencies, following units on a contract or order need not be tested for noise unless otherwise specified (see 6.1).

4.4.2 Pressure and leak tests. The condenser water side and the chilled water side of each unit shall be hydrostatically tested before shipment.

4.4.2.1 Each condenser for type I units shall be tested at 270 psi on the water side. For type II units the condensers shall be hydrostatically tested to 1-1/2 times design submergence. Other assembled components shall be tested for strength and tightness by applying internal pressure equal to 1-1/2 times the design operating pressure (see 3.19). The pressure shall be applied slowly and maintained for 15 minutes. There shall be no leakage at joints or seals or from porosity of castings at any pressure up to 1-1/2 times maximum working pressure.

# 4.4.2.1.1 The void spaces between the double tube sheets of condensers on type II units shall be hydrostatically tested to 1-1/2 times the design submergence pressure after the tests in 4.4.2 and 4.4.2.1 are performed. When the void space is being tested the primary tube sheet shall be examined for leakage. Any leakage shall be corrected.

4.4.2.2 All refrigerant containing parts shall be leak tested at a refrigerant gas pressure equal to 1-1/2 times maximum working pressure or 30 psi whichever is greater. Necessary precautions shall be taken to assure thorough distribution of refrigerant vapor throughout the entire gas volume of parts being leak tested. The leak-detector shall have a leak index sensitivity of 1/2 ounce per year.

4.4.2.3 The entire refrigerant circuit shall be dehydrated and evacuated prior to charging and capacity testing.

4.4.3 Capacity test for type II unit. Each type II unit shall be tested for capacity as required by 4.3.5.

4.4.4 Operation test. Each compressor unit assembly shall be operated for a period of at least 2 hours to demonstrate functional adequacy and proper assembly. Proper operation and adjustment of all controls, purge unit and instruments shall be verified. This test may be run concurrent with other tests.

4.4.5 Maintainability demonstration. The first production unit shall be examined after testing and the capability to maintain, disassemble and repair the unit shall be demonstrated to a Government representative. The unit shall be disassembled, all bearings replaced and reassembled. The controls and instruments shall be removed and replaced at the discretion and direction of the Government representative. The demonstration shall show that access and replacement of components is possible without personnel hazard. The demonstration shall be conducted utilizing the instruction manual, the recommended tools and with other than expert mechanics.

4.5 Test reports. Reports on tests shall be forwarded in triplicate to NAVSEC or the agency concerned to substantiate the fact that the tests have been conducted and the requirements of the specification have been met. Reports shall be complete and assembled in manual form.

# 4.5.1 Reports on capacity tests shall include a summation of the following data:

(a) Compressor:
- Speed of compressor
- Speed of motor
- Compressor suction pressure at inlet
- Compressor suction temperature at inlet
- Compressor discharge pressure at outlet



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Compressor discharge temperature at outlet
Power input to compressor motor
Current input to compressor motor
Voltage at motor terminals
Frequency at motor terminals
Ambient temperature
Barometric pressure

(b) Water chiller:
Refrigerant evaporating temperature
Water temperature at inlet
Water temperature at outlet
Rate of water flow through unit
Water pressure drop through unit

(c) Condenser:
Rate of water flow through unit
Water pressure drop through unit
Water temperature at inlet
Water temperature at outlet

(d) Oil cooler:
Rate of water flow through unit
Water pressure drop through unit
Water temperature at inlet
Water temperature at outlet

4.5.2 Reports on noise and vibration shall be provided in detail or short form as specified (see 6.1).

4.6 Inspection of preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual orders.)

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.

# 5.1.1 Refrigerating units.

# 5.1.1.1 Cleaning. Each equipment shall be cleaned so that all surfaces are free of grease, oil, flux, loose particles and foreign materials. After tests have been completed the oil, water and refrigerant sides shall be drained and cleaned. The refrigerant side of the condenser and cooler shall be cleaned by pumping a cleaning solvent through the unit and a series of filters until no dirt or foreign particles remain in the system. The pumping procedure shall be run for a minimum of 20 minutes through clean filters after it has been indicated that no dirt or foreign particles remain. The water side shall be blown dry with air that is free from oil, water and dirt. Drains shall be open to thoroughly remove any trapped water. The refrigerant side shall be dehydrated by evacuation and the unit provided with a positive pressure charge of dry-oil free nitrogen. The system shall be tagged indicating system is pressure-charged with nitrogen. The lubricating system shall be flushed with normal operating lubricating oil drained and new filters inserted and oil charge renewed.

# 5.1.1.2 Preservation and packaging. Preservation and packaging shall be sufficient to afford protection against corrosion, deterioration and physical damage during shipment from the supply source to the first using activity for immediate installation.

# 5.1.1.3 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during shipment from the supply source to the using activity for immediate installation. The shipping containers and method of packing shall conform to the Uniform Freight or National Motor Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.



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# 5.1.1.3.1 Noise tested units. Equipment which has been noise tested shall be shipped on resilient mounts (see 3.13.4).

# 5.1.1.4 Marking. Shipment marking shall be in accordance with MIL-STD-129. In addition, shipping containers shall indicate that unit is pressure-charged with nitrogen (see 5.1.1.1.)

# 5.1.2 Onboard repair parts and tools. Repair parts shall be preserved, packaged, packed and marked as follows:

(a) Mechanical repair parts. Preservation and packaging level A; packing level C and marking levels A and C respectively in accordance with MIL-R-196.

(b) Electrical. Preservation and packaging level A; packed level C and marked levels A and C respectively in accordance with MIL-E-17555.

(c) Valves, fittings, etc. Valves, fittings, etc., except refrigerant valves, shall be preserved and packaged level A; packed level C and marked levels A and C respectively in accordance with MIL-V-3. Refrigerant valves shall be unit protected by method IA without preservative in accordance with MIL-P-116 packed level C and marked level A and C respectively in accordance with MIL-V-3.

(d) Miscellaneous repair parts:

1. Miscellaneous repair parts such as strainer baskets, capillary tube assemblies, dehydrators, etc., shall be unit protected by method III of MIL-P-116 without use of preservatives. Packing shall be level C and marking levels A and C respectively in accordance with MIL-E-17555.
2. Lubricating oil. Lubricating oil (see 3.31, table I) shall be prepared for shipment in accordance with MIL-STD-290 as follows:

   Capacity - 1-gallon cans conforming to type V, class 4 or 6.
   Packaging - Level A.
   Packing - Level C.
   Marking - In accordance with MIL-STD-290.
3. Refrigerant. In quantity and container specified (see 3.31, table I) shall be packaged, packed and marked in accordance with the Department of Transportation Regulations. In addition, shipment markings shall comply with MIL-STD-129 and color coding of MIL-STD-101.

(e) Tools. Tools shall be preserved and packaged level A; packed level C and marked levels A and C respectively in accordance with MIL-H-15424.

(f) Unless otherwise specified in the contract or order, repair parts and tools shall be individually packaged except when used in sets or quantities greater than one.

# 5.1.2.1 Submarine repair parts. The level A preservation and packaging methods stated in 5.1.2 for repair parts except lubricating oil and refrigerant, shall be modified in accordance with MIL-STD-758 for parts that can be stored onboard ship in bin or drawer type storage.

# 5.1.3 Cushioning. Product cushioning shall be in accordance with the applicable referenced specification, except use of polystyrene "loose-fill" material for packaging and packing applications such as cushioning, filler, dunnage, stuffing, etc., is prohibited.

# 5.1.4 Drawings, manuals and aperture cards. Drawings, manuals and aperture cards shall be prepared for shipment in accordance with the applicable item specification.

# 5.1.4.1 Manuals accompanying equipment. Manuals accompanying equipment (see 3.35) shall be packaged in a greaseproof, waterproof non-fading transparent plastic film and closure shall be heat sealed. Manuals shall not be placed within any flexible sealed barrier enclosing the component. Manuals shall be placed in the shipping container housing the main unit and in a location where they will be readily accessible when the container is opened. The packing list and other shipping documents shall be annotated to indicate which container contains the manual and the approximate location.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).



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5.2.1 The following provides requirements for the various levels of protection during domestic shipment and storage or overseas shipments, which may be required when procurement is made:

# 5.2.1.1 Preservation and packaging.

# 5.2.1.1.1 Level A.

# 5.2.1.1.1.1 Cleaning. Cleaning shall be in accordance with 5.1.1.1.

# 5.2.1.1.1.2 Unit protection. Each complete unit (see 3.3), after cleaning, shall be unit protected in accordance with method IIA of MIL-P-116. All exterior unpainted ferrous metal surfaces shall be coated with a preservative compound conforming to P-19 of MIL-P-116. The flexible barrier bag shall be provided with an external mounted humidity indicator plug. Desiccant quantity shall be in accordance with MIL-P-116. The flexible barrier bag shall be fabricated from material conforming to MIL-F-22191, type I.

# 5.2.1.1.2 Level C. Cleaning, preservation and packaging shall be in accordance with 5.1.1.1 and 5.1.1.2.

# 5.2.1.1.2.1 Equipment shipped by open conveyance shall be shrouded with a reinforced shroud of at least 6 mills thickness to protect the unit(s) from wind blown rain, snow, etc.

# 5.2.1.2 Packing.

# 5.2.1.2.1 Levels A and B. Each complete unit shall be packed in a crate conforming to MIL-C-104, type optional, class 2, style optional. Crates shall be provided with an examination port for viewing of the humidity indicator mounted in the barrier material.

# 5.2.1.2.1.1 Anchoring, blocking, bracing and cushioning shall be in accordance with MIL-STD-1186 and the appendix to the applicable crate specification.

# 5.2.1.2.1.1.1 Use of polystyrene "loose-fill" material for applications such as cushioning, filler, dunnage, stuffing, etc., is prohibited.

# 5.2.1.3 Marking. In addition to any special marking required in the contract or order, shipments shall be marked in accordance with MIL-STD-129 and the appendix to the applicable crate specification.

# 5.2.2 Repair parts and tools. Repair parts shall be packaged level A; packed levels A, B or C and marked in accordance with the requirements of 5.1.2.

# 5.2.3 Drawings, manuals and aperture cards. Drawings, manuals and aperture cards shall be prepared for shipment in accordance with the requirements of 5.1.4.

## 6. NOTES

# 6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number, and date of this specification
(b) Type, class, and arrangement required (see 1.2 and 3.4.1).
(c) Submarine sea water pressure (see 3.1.1, 3.17.2.1, and 3.20).
(d) Refrigerant required if other than R-11 (see 3.2).
(e) Operating requirements (see 3.2.1, 3.2.1.1, 3.30.1 and 3.34.1.1).
    (1) Capacity; tons refrigeration.
    (2) Outlet chilled water temperature.
    (3) Chilled water flow, gpm
    (4) Sea water flow, gpm.
(f) Overall dimensions (see 3.4).
(g) Number of purge units or purge-evacuation units required (see 3.4.3).
(h) Noise and vibration levels and any additional tests required (see 3.12 and 4.4.1).
(i) Whether resilient mounts will be provided by the contractor (see 3.13.5).
(j) Motor service (see 3.16.2 and 3.16.4).
(k) Type motor controller (see 3.16.3).
(l) Pass arrangement if other than that specified for condensers on type I units (see 3.19.2).
(m) Pass arrangement for chiller if other than that specified (see 3.21.3.1).



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(n) Location of chiller head inlet water nozzle if other than that specified (see 3.21.3.2).
(o) Whether onboard repair parts are required and number of sets (see 3.31).
(p) Whether model development drawings are required (see 3.34.6).
(q) Number of manuals required (see 3.35).
(r) Whether shock tests are required (see 4.3.3).
(s) Class of equipment, if other than that specified (see 4.3.3).
(t) Whether shock test foundation design will be provided (see 4.3.3).
(u) Whether equipment which has passed noise tests need be noise tested (see 4.4.1.1).
(v) Whether noise and vibration test reports should be detailed or short (see 4.5.2).
(w) Preservation, packaging and packing requirements if other than those required by 5.1 (see 5.2).

# 6.2 The word "NAVSEC" as used herein refers to the Naval Ship Engineering Center, Propulsion, Power, and Auxiliary Systems Division, Auxiliary Equipment Branch, Washington, D.C. 20360.

# 6.3 The terms "Command", "agency", and "Command or agency" as used herein refer to the procuring activity.

6.4 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
(Project 4130-0139)



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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